

VALUE ADDED PROBLEM-2 (Experiment – 12)

/* Name-Simrat Singh Arora

/PRN-25070521181/

SECTION-D-I

Value Added Problem-2 */

#include <stdio.h>

#include <string.h>

#define MAX 1000000

char stack[MAX]; int

top = -1;

// Push function void

push(char c) {

stack[++top] = c;

}

// Pop function char

pop() { if (top == -

1) return '\0';

return stack[top--];

}

// Check if matching pair int

isMatchingPair(char open, char close) {

if (open == '(' && close == ')') return 1;

if (open == '{' && close == '}') return 1;

```
if (open == '[' && close == ']') return 1;
return 0;
}
```

```
int isBalanced(char *s) {    for
(int i = 0; s[i] != '\0'; i++) {
```

```
// If opening bracket, push    if (s[i] ==
'(' || s[i] == '{' || s[i] == '[') {
push(s[i]);
}
```

```
// If closing bracket
else if (s[i] == ')' || s[i] == '}' || s[i] == ']') {
```

```
if (top == -1) // No matching opening
return 0;
```

```
char topElement = pop();
```

```
if (!isMatchingPair(topElement, s[i]))
return 0;
}
}
```

```
// If stack is empty → balanced    return
(top == -1);
}
```

```
int main() {
```

```
printf("Name
:Simrat Singh
Arora\n");
    printf("PRN
:25070521181\n
");
```

```
printf("Section
:D-1\n");
    printf("Value
Added Problem
: 2\n");
    printf("-----
-----
\n");
char s[MAX];
```

```
printf("Enter expression: ");    scanf("%s",
s);
```

```
if (isBalanced(s))
printf("true\n");    else
printf("false\n");
```

```
return 0;
}
```

main.c

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Run

```
1 /* Name-Simrat Singh Arora
2    /PRN-25070521181/
3    SECTION-D-I
4    Value Added Problem-2 */
5 #include <stdio.h>
6 #include <string.h>
7
8 #define MAX 1000000
9
10 char stack[MAX];
11 int top = -1;
12
13 // Push function
14 void push(char c) {
15     stack[++top] = c;
16 }
17
18 // Pop function
19 char pop() {
20     if (top == -1)
21         return '\0';
22     return stack[top--];
23 }
```

Output

Clear

```
Name :Simrat Singh Arora
PRN :25070521181
Section :D-1
Value Added Problem : 2
-----
Enter expression: {[ ]}
true

=== Code Execution Successful ===
```